Max Begun

Akshay Chebiyam

Samy Kuang Ye

Machine Learning Final Project

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Does the Full Moon Bring More Chaos to New York City?

There is a popular belief that full moons cause people and animals to behave differently. Our group wants to see whether that idea holds up when we look at real data. We plan to study New York City and see if full-moon nights have more noise complaints, public disturbances, reported crimes, or dog bites. We are using the word “chaos” to describe these measurable events, not as an official statistic. New York City is a good setting because it makes large amounts of public data available online.

Our research questions are: (1) Do 311 noise and disturbance complaints increase on or near full-moon nights? (2) Do reported crimes and dog bites show similar patterns? (3) Does knowing the phase of the moon help machine learning models predict unusually chaotic nights after accounting for weather, weekends, holidays, and season? We are interested in whether any relationship exists, including the possibility that there is no meaningful difference.

We plan to combine several public datasets. NYC 311 Service Requests includes complaint types, dates, times, and locations. NYPD Complaint Data provides reported crime dates, locations, and offense categories. The NYC Dog Bite dataset includes reported bites, dates, and boroughs and will be used for a separate exploratory comparison where records are available. We will get full-moon dates and lunar phases from the U.S. Naval Observatory and temperature, precipitation, and cloud cover from the Open-Meteo Historical Weather API. We may also reference FBI Crime Data Explorer statistics for context, but the NYC incident data are better suited to daily comparisons.

For our main machine learning task, each observation will be a borough-night (8:00 p.m. to 4:00 a.m.). We will classify nights as normal (0) or high-chaos (1) using nighttime 311 noise and disturbance complaints. High-chaos will mean a complaint count above the borough-specific 75th percentile, calculated from training data only. Predictors include moon phase, illumination, weekday, holidays, month, weather, borough, and complaints from previous nights. We plan to study 2020 through 2025, depending on data coverage. Crimes and dog bites are separate comparison outcomes, not same-night predictors.

We will clean and join datasets by date and borough, address missing values, and graph patterns across moon phases. We will compare a majority-class baseline with decision tree, random forest, KNN, and gradient boosting classifiers. After tuning on training data, we will evaluate predictions on later dates using precision, recall, F1-score, ROC-AUC, and confusion matrices. We will compare models with and without lunar variables to determine whether moon phase improves prediction. No information from the night being predicted will be used as an input.

Our schedule is to confirm data availability and collect sample records during November 2–8; clean and combine the datasets November 9–15; explore patterns and produce visualizations November 16–22; train initial classifiers November 23–29; and tune and evaluate them November 30–December 6. From December 7–11, we will finish the notebook, executive summary, presentation slides, and five-to-ten-minute video. All three team members will contribute to the data work, modeling, analysis, and presentation.

One limitation is that complaints and crime reports only capture incidents that get reported. Differences in reporting, weather, holidays, and neighborhood activity may affect our results. Even if we find an association with the full moon, we will not be able to conclude that the moon caused those events.